

Methane Low Cost Sensor (MLCS): 2026 milestone review

Written by:

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Contributors:

Vincent Valentin Scholz, Jonas Stage Sørensen, Bruce Linquist, Alison Hoyt, Benjamin Runkle, Scott Fendorf, Sebastian Echeverria, and members of the Linquist, Runkle, and Hoyt groups who have offered their time, expertise, and passion.

A note of gratitude

Whilst the findings thus far are not picture perfect, they are however, a testament to the dedication of many amazing individuals. Each data point shown in a plot represented tremendous effort - be it driving out to a field, setting up a greenhouse trial, 3D printing chambers, running samples with the chromatograph, carrying a Li-Cor 7810, sensor deployment, or the inevitable data crunching once all analytical work had been completed. A most heartfelt thank you to every single person who gave their time, energy, and expertise to this project.

*So they said yes,
And a word turned into seconds and days.*

*Point by point,
Date by date,
A picture came to life.*

*Of voltages and resistances,
Soils and puffs of gas.
For a crop that feeds the world,
And a dream to grow it,
Perhaps a little more gently, on our planet.*

~ Thank you all, for a labor of love

1. Overview

Low-cost metal oxide semiconductor methane sensors are gaining attention as a cheap way to measure ecosystem GHG fluxes. A methane low-cost sensor (MLCS) was manufactured by Jonas Stage Sørensen (MOS; NGM2611-E13, Figaro, USA) and validated in a Lake system (Sørensen et al., 2024, <https://doi.org/10.1029/2024JG008035>). Vincent Scholz proposed adapting the MLCS as an alternative and more accessible method to measure CH₄ flux in rice systems. This milestone report is meant to be a concise summary of all data analysed for all collaborators in this project.

The overall aim of the study was to:

- (a) Design MLCS measurement protocols specific to rice systems
- (b) Evaluate the accuracy of using MLCS for field CH₄ measurements compared to established methods, including gas chromatography (GC) and Li-Cor 7810 (Li-Cor).

Overall, the study has demonstrated the following:

- (a) The MLCS can be utilised for CH₄ measurements in rice using static chambers with air circulation and appropriate deployment duration with minimal disturbance (>20 mins). This was supported by the good correlation of MLCS and GC data (change in CH₄ concentration over time).
- (b) To obtain accurate values, the MLCS must not be deployed in a static chamber for a short period of time. Instead, a prolonged duration (>20 mins) is needed. This was supported by the lack of agreement between Li-Cor and GC data, where a short chamber enclosure of 3 mins was used.
- (c) We propose that the MLCS can be easily used for low-cost measurements with appropriate calibration and deployment.

All data and scripts used for analyses are publicly available in a data repository: https://github.com/XiaoZhangZhangRice/Methane_Insight

2. MLCS deployment and data extraction

In almost all analyses in this report, the change in CH₄ concentration over time (ΔC , ppm/hr) was used as a comparison between methods. ΔC (ppm/hr) was chosen as a response variable as it is mathematically associated with flux (g CH₄ ha⁻¹ day⁻¹) and allowed the least computation with covariates (e.g. chamber volume).

The MLCS was warmed up for about 1 hr before deployments and was attached to the inside of a static chamber. For the GC method, the enclosure period was approximately 75 mins. For the LiCor method, the enclosure period was 3 minutes.

ΔC was extracted from the MLCS using the output CSV files and an online Shiny app, where sensor-specific calibrations were included in the web app (<https://methaneinsight.shinyapps.io/fluxR/>).

3. Field validation of MLCS using gas chromatography methodology (California)

Three unique MLCS were deployed to 9 unique plots for a season-long gas sampling campaign during summer 2024. For each given deployment, change in CH₄ concentration over time (ΔC , ppm/hr) was determined concurrently using (a) gas chromatography (GC, Shimadzu GC-2014), and (b) MLCS in combination with an online R shiny app (<https://methaneinsight.shinyapps.io/fluxR/>).

Gas samples for GC analysis were taken at times 0, 21, 42, and 63 minutes after chamber enclosure. Before samples were taken, a fan was turned on for approximately one minute to mix the air. ΔC_{GC} was determined as the slope from linear regression of CH₄ concentration against time.

The MLCS was allowed to warm up for an hour before deployment and fixed to the top of a GHG chamber before chamber closure. A CSV file was generated for each deployment and was subsequently loaded into the R shiny app to determine ΔC_{MLCS} . During initial deployment, it was discovered that fan operations caused rapid drops in CH₄ concentrations (Fig 1). Consequently, ΔC_{MLCS} was determined using 3 different methods (Fig 1).

- (1) Fan off: ΔC_{MLCS} estimated using a region where no fan was turned on and there was a linear relationship between CH₄ (ppm) and time.
- (2) Fan on: ΔC_{MLCS} was estimated from the end of a sampling session where the fan was left on for approximately 7 minutes. The region used had a linear

relationship between CH₄ (ppm) and time. Note: Switching on a fan at the end of a sampling session only started after 19 Jul 2024 after it was discovered that fan operation caused rapid drops in CH₄ concentration.

- (3) Three point: CH₄ concentrations were taken from 3 points where the fan was not in operation along with the associated timestamp. ΔC_{MLCS} was determined as the slope of a 3-point linear regression.

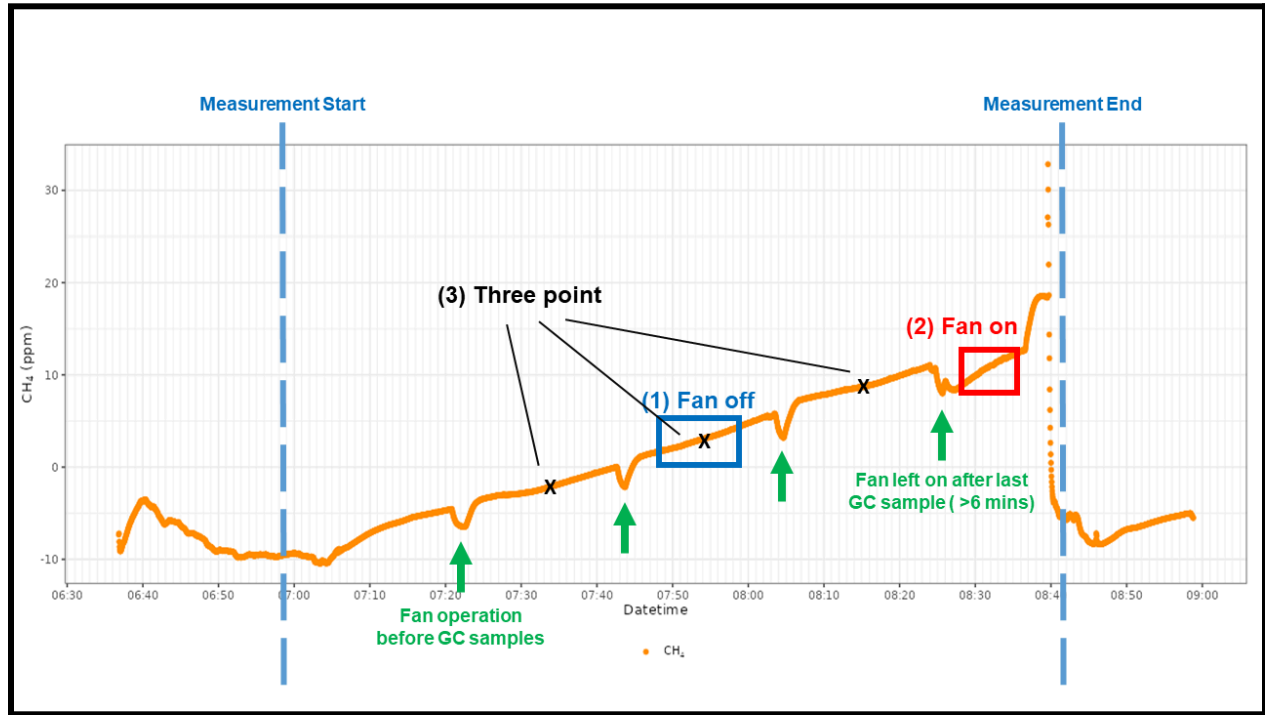


Fig 1. A typical MLCS R shiny output showing change in CH₄ concentration over time. Large drops in concentration were observed when the fan was turned on (green arrows). ΔC_{MLCS} was predicted in 3 ways: (1) fan off (blue box), (2) fan on (red box), and (3) three-point (black crosses).

All MLCS and GC determined ΔC values were plotted (Fig 2 and 3). Overall, ΔC_{MLCS} determined using the fan on method (Fig 3) was the closest to ΔC_{GC} - points were the closest to the 1:1 line. This might be related to the calibration conditions of the MLCS, whereby there was active airflow.

However, for the fan off and three-point methods, ΔC_{GC} was almost always higher than ΔC_{MLCS} - evidenced by a large number of points above the 1:1 line (Fig 2a and d). Based on the linear regression, ΔC_{GC} was approximately two times larger than ΔC_{MLCS} for these two methods. However, when raw ΔC_{MLCS} values were transformed using the linear model, there was excellent agreement between ΔC_{GC} and predicted ΔC_{MLCS} (Fig 2b and d) - evidenced by the high correlation value. The majority of predicted ΔC_{MLCS} values also have a small difference compared ΔC_{GC} and the majority of differences lie close to the median difference value (Fig 2c and f). Mathematical transformations are common in the application of low-cost sensors, and appear to be a sensible way forward as there is a good prediction ability and eliminate potential inaccuracies caused by fan operation.

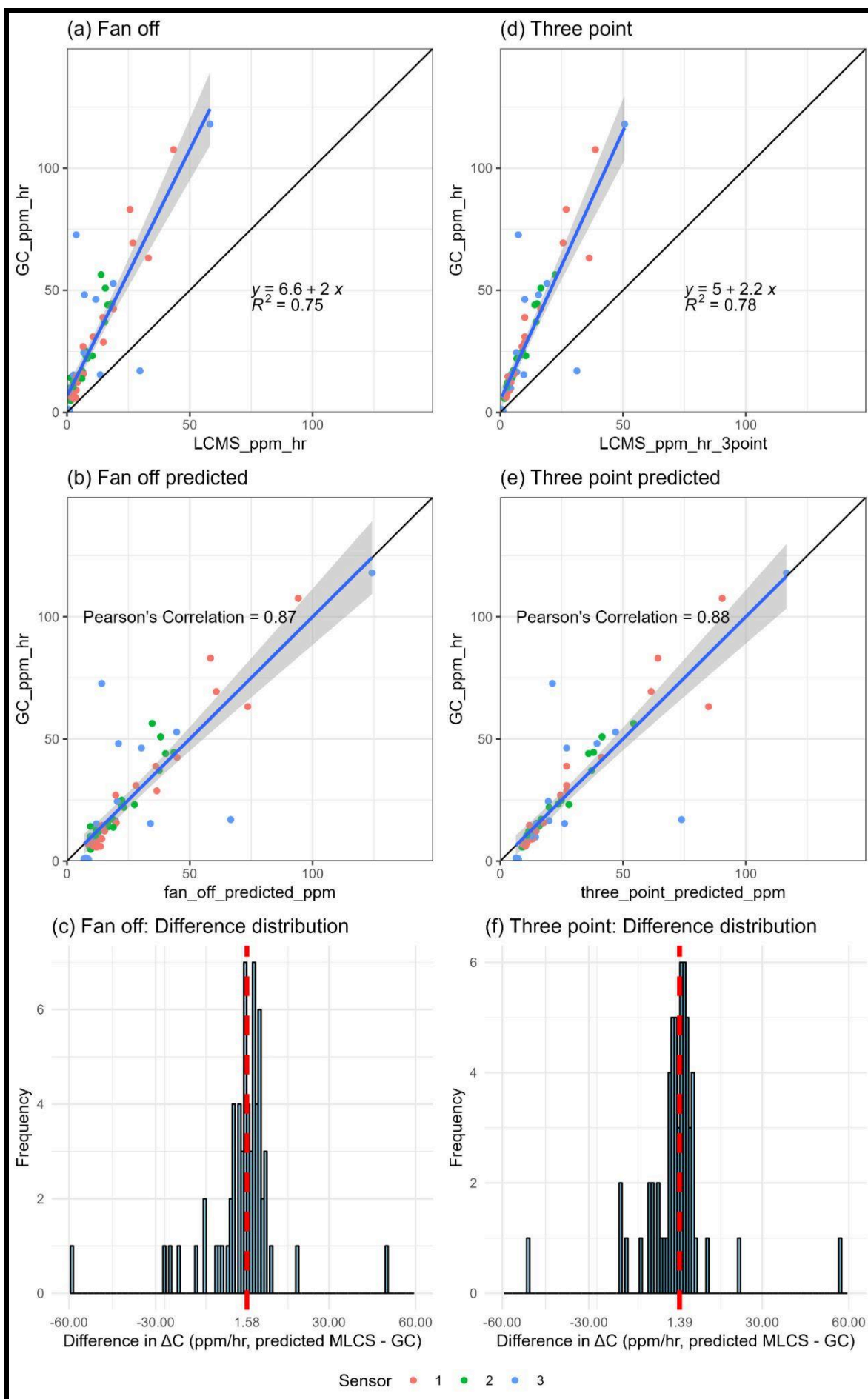


Fig 2. Linear regression, 2a and d, of CH₄ concentration change (ΔC , ppm/hr) for gas chromatography (GC) measured and MLCS measured values using the fan off and three-point methods. The black line represents a 1:1 relationship (i.e. $y=x$). The equation of the line of best fit and R^2 are displayed. Raw ΔC_{MLCS} values were transformed by the linear regressions shown in 2a and d (predicted values), and are shown in the middle two rows (2b and e). The shaded area represents the 95% confidence interval. 2c and f are histograms of differences between predicted MLCS and GC ΔC measurements. The median value is indicated by a red vertical dashed line.

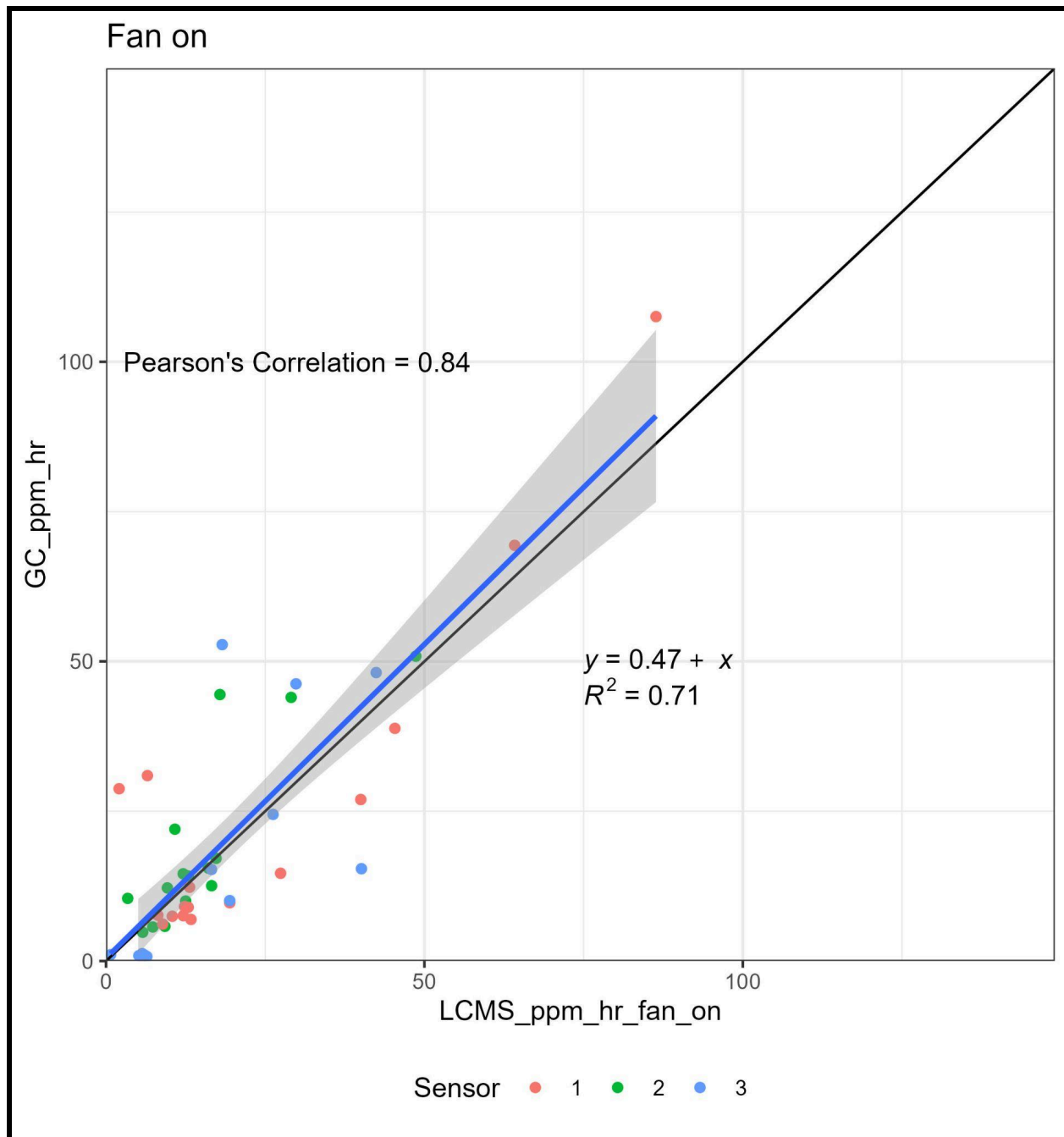


Fig 3. Linear regression of CH₄ concentration change (ΔC , ppm/hr) for gas chromatography (GC) measured and MLCS measured values using the fan-on method. The black line represents a 1:1 relationship (i.e. $y=x$). The equation of the line of best fit and R^2 are displayed. The shaded area represents the 95% confidence interval.

4. Field validation of MLCS using Li-Cor 7810 methodology (Arkansas)

One MLCS was deployed in Arkansas. For each deployment, (ΔC , ppm/hr) was determined concurrently using a (a) trace gas analyser (Li-Cor 7810, referred to from hereon as Li-Cor), and (b) MLCS.

For a given deployment, the Li-Cor (<https://www.licor.com/products/trace-gas/LI-7810>) was connected to a smart chamber (<https://www.licor.com/products/soil-flux/smart-chamber>), with the MLCS attached to the inside of the chamber. ΔC was measured for 3 mins using both instruments. In 2024, two measurements were taken for each collar (or more generally, biological replicate). After 3 minutes, the smart chamber was vented and closed again, giving two replicates for each collar. As such, we also considered two measurements from the MLCS. In 2025, only one measurement was taken from each collar (i.e. no replicates).

$\Delta C_{\text{Li-Cor}}$ was determined using a typical deadband method. The first 30 seconds of each measurement were removed, and subsequent points were used to determine the slope $\Delta C_{\text{Li-Cor}}$ and R^2 . For quality control, only $\Delta C_{\text{Li-Cor}}$ values with R^2 values greater than 0.9 were retained for further analyses.

The raw data file from the Li-Cor was also used to create a base dataframe for obtaining ΔC_{MLCS} values using unique identifiers of field, collar, replicate, and date, and arranged in chronological order ($n = 558$). The MLCS datalog was sorted by day and processed using the shiny app one by one. Measurement events were compared to this constructed dataframe with aid from the H₂O lines to make sure measurements “lined-up” (Fig. 4a). Because there is time drift between the Li-Cor and MLCS, it was impractical to use timestamps for data processing. A relatively linear portion (30-60s) of the CH₄ line from the end of each measurement was used to calculate ΔC_{MLCS} (Figure 4b and 4c). Reti manually extracted ΔC_{MLCS} values from July to November 2024, while Zhang manually extracted ΔC_{MLCS} values from December to May 2025. Zhang and Reti also rated about 20 values from each other’s periods and did not find big deviations in the values obtained. ΔC_{MLCS} values from June 2025 were excluded from the analyses as there were reports of device malfunction.

The ΔC_{MLCS} dataset was merged with the $\Delta C_{\text{Li-Cor}}$ using the dplyr package and unique identifiers (field, date, collar, replicate). The final dataset totalled 124 observations after accounting for quality control (exclusion of observations where R^2 used to $\Delta C_{\text{Li-Cor}}$ was lower than 0.9).

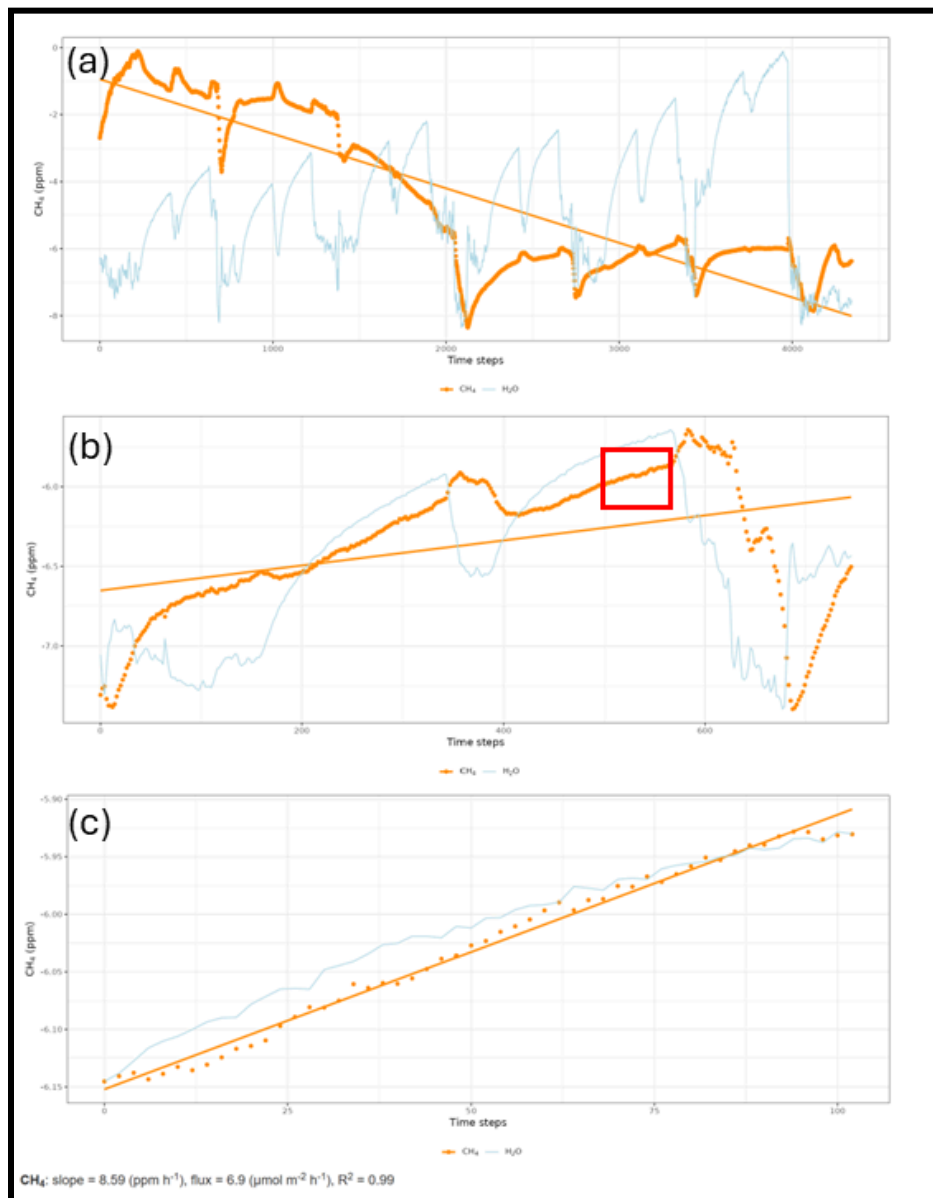


Fig 4. Shiny app workflow for Arkansas study. (a) H₂O lines (blue) were used to identify measurements (each peak is one measurement), (b) a linear portion of the CH₄ line (orange), usually towards the end of the measurement, was (c) used to calculate ΔC_{MLCS} (ppm/hr, bottom left of figure).

In general, there was no relationship found between ΔC_{MLCS} and $\Delta C_{\text{Li-Cor}}$ (Fig. 5a). The data were investigated for any periodic error. First, observations were sorted by month and year (Fig. 5b). Visually, it was seen that the Aug 2024 data had wide variability, and other values tended to lie close to the X-axis ($y=0$) (Fig. 5b), suggesting a trend of random error. Additionally, we evaluated the differences between ΔC_{MLCS} and $\Delta C_{\text{Li-Cor}}$ at each sampling point (Fig. 6). Once again, large variability was observed within the dataset on each sampling day, with variability increasing as $\Delta C_{\text{Li-Cor}}$ increases. There were no periodic trends of over- or underestimation across time, suggesting random errors.

Additional covariates, such as temperature and relative humidity, were used in the MLCS to obtain CH₄ concentration values. These two covariates were investigated and also showed no signs of additional concerns (Fig. 7).

K-means clustering was also performed to evaluate if there were any clusters that were not immediately visually observable (Fig. 5c). Booth 3 and 5 centre approaches were used, but only 5 was shown here. There appears to be no direct relationship between the cluster locations and when the data was collected.

Lastly, Reti, Zhang, and Vincent visually observed that the second replicate of each collar appeared more linear (2024 data only). Thus, we retained only the second replicate. Even when excluding single or first replicates, there appears to be no apparent correlation between $\Delta C_{\text{Li-Cor}}$ and ΔC_{MLCS} .

We hypothesise that this observation is because of (1) the short duration of chamber closure and (2) disturbance from the sampling method. The Li-Cor smart chamber was closed for a short duration of 3 mins. Additionally, the chamber has a fan included, which can affect the operating temperature of the metal oxide sensor due to air movement. Together, this sampling method presents a short time frame characterised by an initial deadband due to pressure differences and subsequent noise with additional air movement, resulting in erroneous measurements. From Jonas's publication, the results suggest that for ΔC_{MLCS} to stabilise, the instrument needs to be in a chamber for more than 20 mins (Fig. 8ab) to obtain a good measurement. From the California study, we also observed a stabilisation period of about 5 mins after the fan was turned on for the "fan-on" segment. As such, the lack of correlation between ΔC_{MLCS} and $\Delta C_{\text{Li-Cor}}$ was likely due to the short chamber enclosure, resulting in inaccurate measurements from the MLCS.

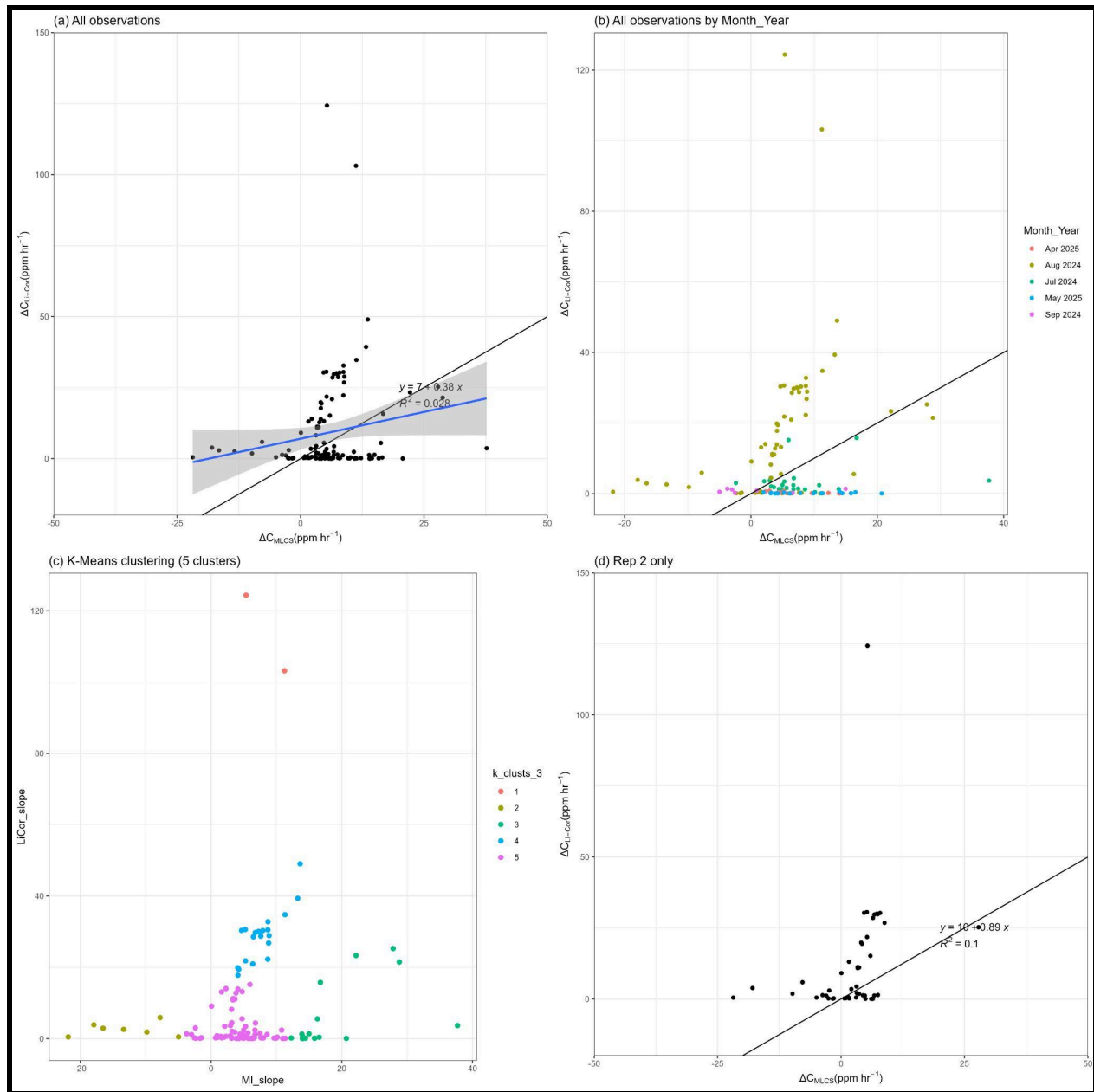


Fig. 5. Comparison of ΔC_{MLCS} and ΔC_{Li-Cor} values. For all graphs, the black line represents a 1:1 relationship ($y=x$). (a) All observations plotted with linear regression (95% confidence interval), (b) observations grouped by month and year, (c) K-means clustering with 5 centers, and (d) replicate 2 observations only.

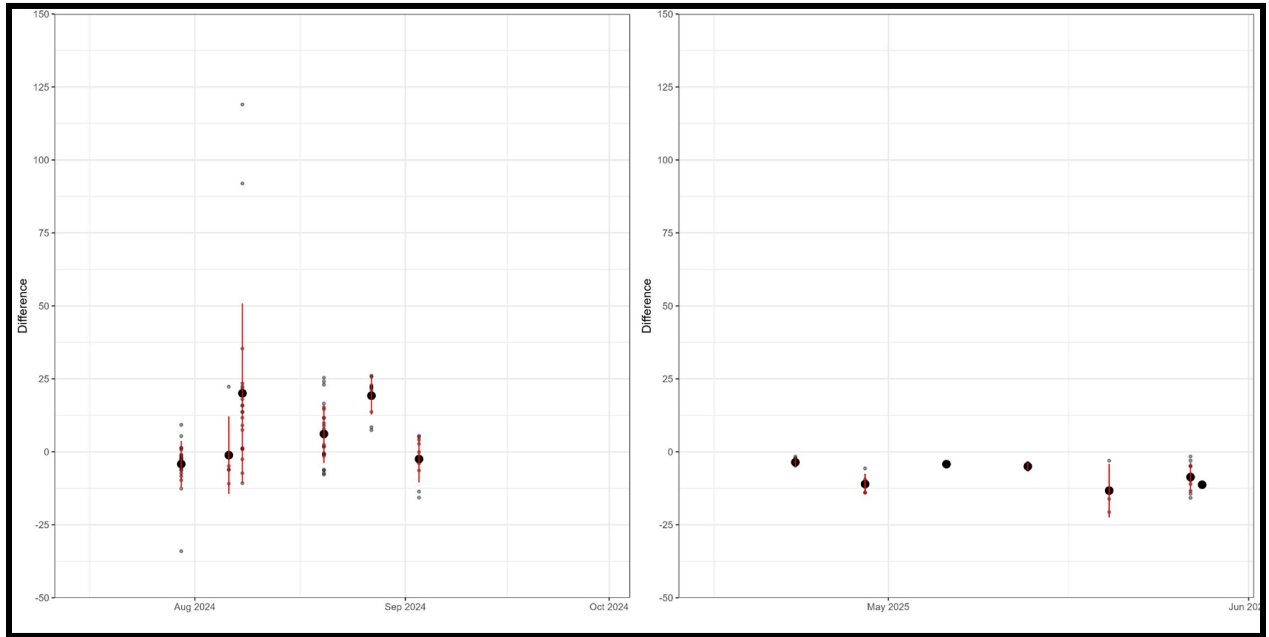


Fig. 6: Difference between ΔC_{MLCS} and $\Delta C_{\text{Li-Cor}}$ values for each sampling date. Large black dots represent the mean and small grey dots represent each individual observation. Red lines represent the standard deviation.

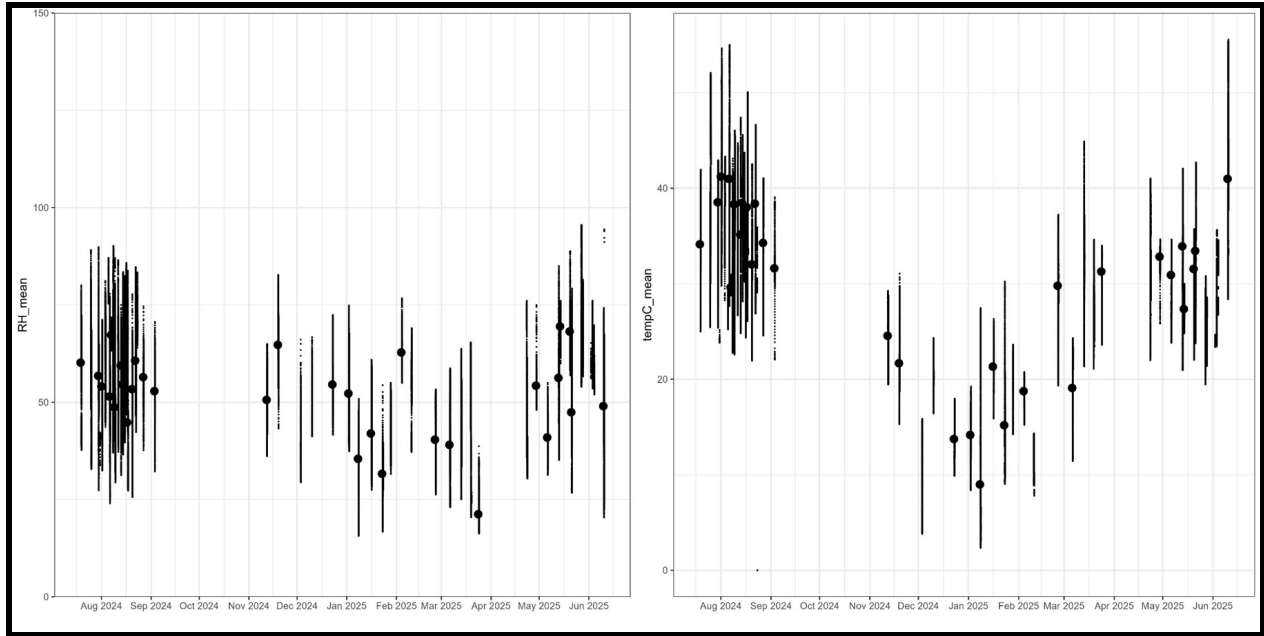


Fig. 7: Distribution of covariates, relative humidity (RH, %; left) and temperature (Temp_C, °C; right) for each sampling date. Values were obtained from the MLCS log. Large black dots represent the mean, and small grey dots represent each individual observation. Black lines represent the standard deviation.

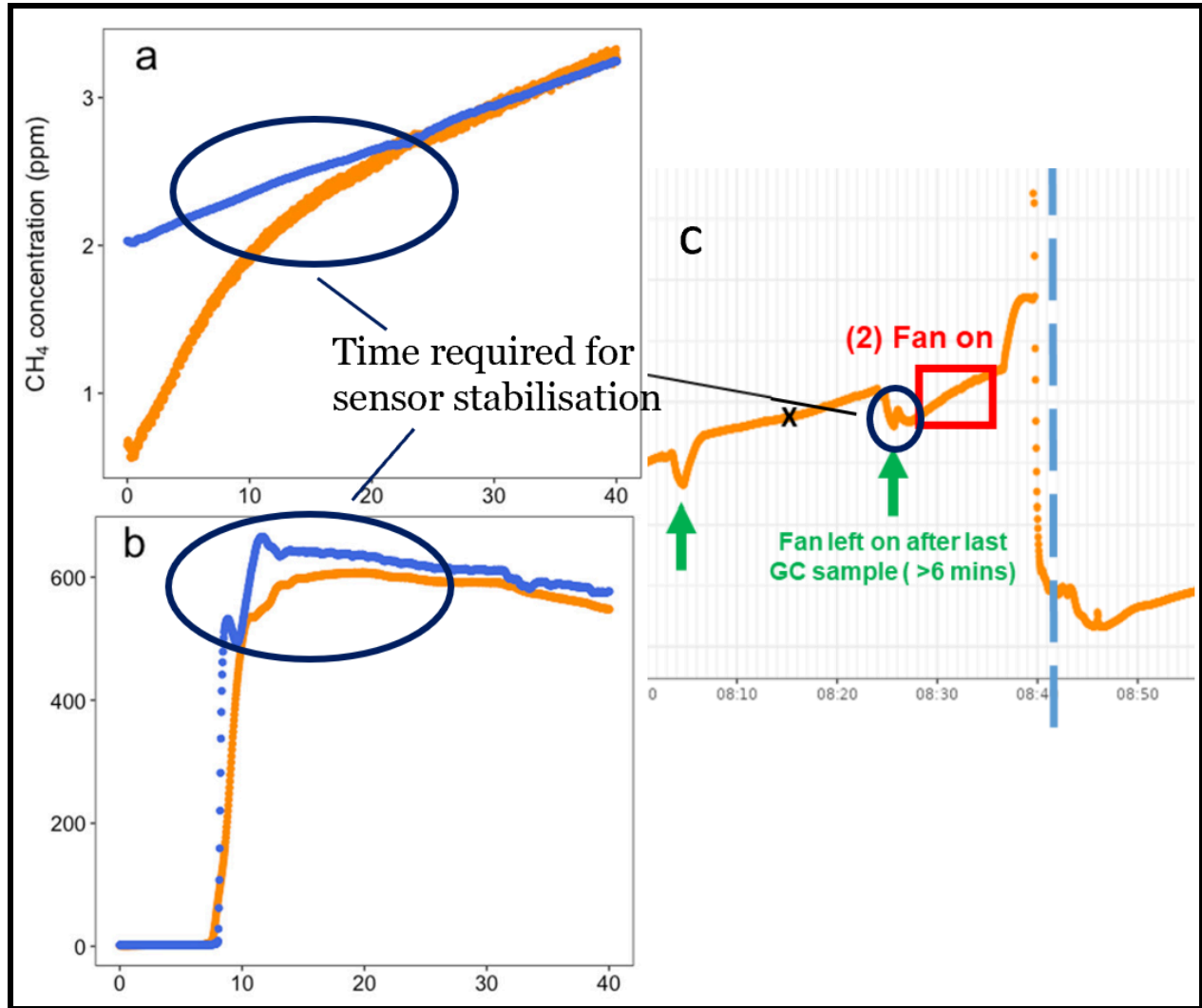


Fig. 8. Comparisons of ΔC from Jonas's publication (a: low-flux scenarios; b: bubble event) and the California experiment (c: fan-on segment), showing time required for ΔC_{MLCS} to stabilise.

5. Conclusions and future directions

Based on the results of the California field study, we conclude that the MLCS has the potential to be developed for field measurements of CH₄ concentration change in rice. Importantly, there was a good correlation between ΔC_{MLCS} and ΔC_{GC} (Fig 3) for the fan-on method. Additionally, when raw ΔC_{MLCS} values obtained from three-point and fan-off methods were transformed, there was excellent agreement between ΔC_{MLCS} and ΔC_{GC} (Fig 2).

Based on the results of the Arkansas field study, we show that a short chamber enclosure period of 3 mins was insufficient to obtain a good reading from the MLCS, resulting in no correlation between ΔC_{MLCS} and $\Delta C_{\text{Li-Cor}}$ (Fig. 5). This negative result is important as it is an important methodological consideration for future users and measurements.

To conclude, the application of Figaro's low-cost methane sensor in rice field methane emissions is feasible. The experiment was an important first step in troubleshooting hardware, and identifying calibration and deployment challenges, providing a clear path forward for improving sensor precision and protocol development.

In one sentence: The sensor works well, but needs to be placed in a chamber under stable conditions with air circulation for a prolonged period (>20 mins) to obtain good measurements.

Climate change is real, and we truly hope that we can continue to work towards our overall mission of using MLCS to make GHG flux measurements more financially and operationally accessible for researchers, legislators, farmers, and other stakeholders across the global rice industry.